Midterm Practice Questions · Sessions 2 to 6

MATH60033A · Quantitative Methods in International Business

Thierry Warin, PhD · HEC Montréal · A2026

These questions train for the midterm under its conditions: **2 hours 30 minutes, on paper, in pen, with a non-programmable calculator, and nothing else**. The midterm tests your **understanding** — what a method does, what a number means, and what it does not license — not your ability to carry out long calculations. The practice follows the exam's six parts:

Part	What it asks	Here
I — Europe 2031	answer in words: what AI the scenario is about, and Europe's options	9 prompts
II — The readings	research question, model, critiques and an alternative model, for each paper	5 readings
A — multiple choice	one correct answer out of four	40 questions
B — short answers	state a concept precisely, in two or three sentences	21 questions
C — reading output	interpret an output in words, or find the planted error	8 items
D — judgement	say what a number does *not* license	5 prompts

To print: questions · answer key (PDF).

Each question carries its session and its difficulty in brackets, so you can train one session at a time, and build up from easy to difficult:

- **easy** (25 questions) — recall a definition, or recognise the right reading of a result;
- **medium** (44) — combine two ideas, or explain why a tempting reading is wrong;
- **difficult** (19) — a subtle distinction, an argued position, or several faults to find at once.

How to use it. Answer first, on paper, without notes. Only then open the answer key. An answer you recognise is not an answer you can produce; the exam asks you to produce it. The midterm summary holds the ideas you are expected to carry in your head.

Part I — Europe 2031

Answer in words, from the scenario and our Session 1 conversation, "what AI actually is". No calculation is needed. Half a page each.

I1. *(S1 · easy) Europe 2031* is a **scenario**, not a **forecast**. What is the difference? Where does the fact-based part of the text end and the speculative part begin, and why does that seam matter for anyone who quotes it?

I2. *(S1 · medium)* In Session 1 you downloaded a language model of a few gigabytes. What is in that file, and what computation runs when it answers a question? Where does its competence come from, and where does its stop?

I3. *(S1 · medium)* The scenario measures a region's AI power in **gigawatts of compute** (255.1 GW for the United States in 2031, 20.9 GW for Europe). What do these numbers represent, and what does the choice of that unit assume about what makes AI powerful?

I4. *(S1 · medium)* Compute, data and mathematics (the algorithms) are the three ingredients of an AI system. Explain each in your own words, and say on which of the three Europe is best placed to compete.

I5. *(S1 · medium)* What is **vertical AI**? Give two examples in European industry, and say whether vertical AI is a way for Europe to escape the frontier race the scenario describes.

I6. *(S1 · medium)* What is a **small language model**? Give two advantages and two limits for a European firm that chooses one over a frontier model run by a US provider.

I7. *(S1 · medium)* Why does the scenario propose **Special Compute Zones** — faster permits, dedicated teams, grid connections from retired power plants? What could go wrong with a strategy built around data centres?

I8. *(S1 · difficult)* The scenario contrasts **compute sovereignty** (Europe builds and controls its own AI stack) with **governed interdependence** (Europe works with US providers but anchors critical infrastructure under European law). Which do you find more realistic, and why?

I9. *(S1 · difficult)* "What in this course is AI, and what is a hundred-year-old statistical method with a new name?" Answer the question for the methods of sessions 2 to 5, and say whether the distinction matters.

Part II — The readings

*For each required reading, answer the same four questions: **(a)** the research question; **(b)** the model — the outcome, the unit of observation, the explanatory variables and the method; **(c)** two distinct critiques a referee could raise; **(d)** another model from this course that could answer the question, and what it would report.*

II1. *(S2 · medium)* **Fraiberger, Lee, Puy and Sahay (2021)**, *Media sentiment and international asset prices.*

II2. *(S3 · medium)* **Amsili, van Es and Schindelbeck (2024)**, *Pedotransfer functions for field capacity, permanent wilting point, and available water capacity.*

II3. *(S4 · difficult)* **Saganowski et al. (2019)**, *Analysis of group evolution prediction in complex networks.*

II4. *(S5 · medium)* **Blonigen and Piger (2014)**, *Determinants of foreign direct investment.*

II5. *(S6 · difficult)* **Topalova and Khandelwal (2011)**, *Trade liberalization and firm productivity.*

Part A — Multiple choice

Tick one answer. On the exam, each correct answer earns the same marks and a wrong answer costs nothing.

Session 2 · Describing a variable

A1. *(S2 · medium)* The sample variance divides by $n-1$ rather than n because:

- **A.** it makes the variance larger, which is safer
- **B.** the formula would otherwise not have the units of x^2
- **C.** the deviations from $\bar{x}$ sum to zero, so only $n-1$ of them are free
- **D.** one observation is always an outlier

A2. *(S2 · easy)* A 5% trimmed mean is:

- **A.** the mean of the 5% largest values
- **B.** the mean computed after dropping 5% of the observations from each tail
- **C.** the median plus 5%
- **D.** the mean multiplied by 0.95

A3. *(S2 · easy)* Household income is strongly right-skewed. To describe a "typical" household, you report:

- **A.** the median
- **B.** the mean
- **C.** the maximum
- **D.** the variance

A4. *(S2 · medium)* The 68–95–99.7 rule may be applied to a variable that is:

- **A.** strictly positive
- **B.** symmetric, whatever its tails
- **C.** measured in percent
- **D.** roughly normal

A5. *(S2 · medium)* A variable's skewness is -1.4, well beyond its threshold. Then:

- **A.** it is roughly normal
- **B.** its mean is above its median
- **C.** it has a long left tail, and its mean is below its median
- **D.** it has no extreme values

Session 3 · Can the regression be trusted?

A6. *(S3 · easy)* Ordinary least squares chooses the line that:

- **A.** minimises the sum of the squared residuals
- **B.** passes through the largest number of points
- **C.** minimises the largest residual
- **D.** maximises the slope

A7. *(S3 · easy)* A residual is:

- **A.** the slope of the line
- **B.** the predicted value
- **C.** the average of y
- **D.** the observed value minus the fitted value

A8. *(S3 · difficult)* In a multiple regression, the coefficient on x_j measures the relationship between y and:

- **A.** x_j, ignoring the other regressors
- **B.** the part of x_j that the other regressors do not explain
- **C.** the sum of all the regressors
- **D.** x_j in the observations where the other regressors are zero

A9. *(S3 · difficult)* For OLS to be unbiased, which of these is **not** needed?

- **A.** the model is linear in its parameters
- **B.** the errors have mean zero given the regressors
- **C.** the errors have constant variance
- **D.** no regressor is an exact combination of the others

A10. *(S3 · easy)* A regression of GDP per capita on a productivity index has an intercept of −€86,000. The index runs from 72 to 129 in the data. The intercept should be:

- **A.** left uninterpreted: it is a prediction at an index of 0, far outside the data
- **B.** reported as the income of the least productive country
- **C.** read as proof that the model is wrong
- **D.** removed from the model

A11. *(S3 · medium)* A regressor has a variance inflation factor of 15. This means:

- **A.** its coefficient is 15 times too large
- **B.** it explains 15% of y
- **C.** it should be multiplied by 15
- **D.** it is almost a combination of the other regressors, so its coefficient is unstable

A12. *(S3 · medium)* An observation's **leverage** depends on:

- **A.** its residual only
- **B.** how unusual its values of the regressors are
- **C.** its value of y only
- **D.** the number of regressors only

A13. *(S3 · easy)* Cook's distance flags one country-year far above all the others. You:

- **A.** delete it and report the new fit only
- **B.** ignore it: one row cannot matter
- **C.** find out which row it is and why, and report the fit with and without it
- **D.** add a dummy for it and say nothing

A14. *(S3 · medium)* AIC and BIC disagree: AIC prefers the larger model, BIC the smaller one. The usual reason is that:

- **A.** BIC charges more for each extra parameter once the sample is not tiny
- **B.** AIC is computed on different data
- **C.** BIC ignores the fit of the model
- **D.** one of them must be miscalculated

A15. *(S3 · easy)* Two models may be compared on AIC only if they are:

- **A.** both significant

- **B.** both linear
- **C.** both estimated by the same person
- **D.** fitted on the same rows

A16. *(S3 · medium)* A firm's management quality raises its exports **and** is higher in firms that invest more. A regression of exports on investment alone will:

- **A.** be unbiased
- **B.** overstate the effect of investment
- **C.** understate the effect of investment
- **D.** always give a negative coefficient

A17. *(S3 · easy)* More educated regions have higher wages. Which sentence is licensed?

- **A.** "Education causes higher wages."
- **B.** "Raising education will raise wages."
- **C.** "Education is associated with higher wages."
- **D.** "Wages do not affect education."

Session 4 · Categorical outcomes

A18. *(S4 · easy)* A probability of 0.25 corresponds to odds of:

- **A.** 1 to 3, about 0.33
- **B.** 0.25
- **C.** 3 to 1
- **D.** 0.75

A19. *(S4 · medium)* Odds of 4 (four to one) correspond to a probability of:

- **A.** 0.25
- **B.** 4
- **C.** 0.40
- **D.** 0.80

A20. *(S4 · easy)* An odds ratio of 0.70 for a training programme means that the programme:

- **A.** lowers the probability by 70%
- **B.** multiplies the odds by 0.70 — 30% lower odds
- **C.** has no effect
- **D.** lowers the probability by 0.70 points

A21. *(S4 · easy)* A logistic regression is estimated by:

- **A.** least squares on the 0/1 outcome
- **B.** soft-thresholding
- **C.** maximum likelihood
- **D.** cross-validation

A22. *(S4 · medium)* You want to test whether four region dummies matter **jointly** in a logit. You use:

- **A.** a likelihood-ratio test comparing the models with and without them
- **B.** the largest of the four z-statistics
- **C.** the R^2
- **D.** the odds ratio of the first dummy

A23. *(S4 · medium)* A logit has a McFadden pseudo-R^2 of 0.25. This means:

- **A.** the model explains 25% of the variance of the outcome
- **B.** 25% of the predictions are correct
- **C.** the model is useless
- **D.** a reasonable improvement over the intercept-only model; it is not a share of variance

A24. *(S6 · medium)* Export values are strongly right-skewed. Before regressing them on firm size, an analyst takes their logarithm. This:

- **A.** removes every outlier
- **B.** pulls in the long tail, and a coefficient then reads approximately as a percentage change
- **C.** makes the regression causal

- **D.** is forbidden in a linear regression

A25. *(S4 · medium)* An ordinal logit for four ordered categories has:

- **A.** four cutpoints and four coefficients per predictor
- **B.** no cutpoints
- **C.** three cutpoints and one coefficient per predictor
- **D.** one equation per category against a baseline

A26. *(S4 · difficult)* The **proportional-odds** assumption says that:

- **A.** each predictor has the same odds ratio at every cut between categories
- **B.** every category has the same probability
- **C.** the odds always equal the probability
- **D.** the cutpoints are equally spaced

A27. *(S4 · difficult)* In a multinomial logit, you change the baseline category from *academic* to *general*. Then:

- **A.** the predicted probabilities change
- **B.** the model can no longer be estimated
- **C.** nothing changes at all
- **D.** the coefficients change meaning, but the predicted probabilities stay the same

A28. *(S4 · easy)* **Precision** is:

- **A.** the share of all cases classified correctly
- **B.** among the cases flagged, the share that are truly positive
- **C.** among the true positives, the share that are flagged
- **D.** the area under the ROC curve

A29. *(S4 · easy)* **Recall** is:

- **A.** the share of all cases classified correctly
- **B.** among the cases flagged, the share that are truly positive
- **C.** among the true positives, the share that are flagged
- **D.** the threshold used

A30. *(S4 · medium)* A model is **well calibrated** when:

- **A.** among the cases it gives a probability of 20%, about 20% are positive
- **B.** it ranks every positive above every negative
- **C.** its accuracy exceeds 90%
- **D.** it uses a threshold of 0.5

A31. *(S4 · difficult)* A default model has an AUC of 0.93, but its predicted probabilities are all about twice the observed default rates. This is:

- **A.** impossible
- **B.** a sign of a coding error
- **C.** a perfectly calibrated model
- **D.** possible: it ranks well (discrimination) but its probabilities are wrong (calibration)

A32. *(S4 · medium)* Missing a default costs a bank far more than a false alarm. Compared with 0.5, the threshold for flagging a borrower should be:

- **A.** higher
- **B.** lower
- **C.** the same: 0.5 is always optimal
- **D.** set to the base rate whatever the costs

Session 5 · Regularisation

A33. *(S5 · medium)* Ridge regression can predict new data better than OLS because:

- **A.** it is unbiased
- **B.** it uses more data
- **C.** it accepts a little bias for a large cut in variance
- **D.** it removes every correlated predictor

A34. *(S5 · easy)* As the ridge penalty λ grows very large, the coefficients:

- **A.** all shrink toward zero, and the model predicts the mean
- **B.** become the OLS coefficients
- **C.** grow without limit
- **D.** become exactly zero one at a time, like the lasso

A35. *(S5 · difficult)* With more predictors than observations, OLS has no unique solution but ridge does, because:

- **A.** ridge drops half of the predictors
- **B.** ridge uses the median instead of the mean
- **C.** ridge needs no data
- **D.** adding the penalty makes the problem solvable for any $\lambda > 0$

A36. *(S5 · easy)* Predictors must be standardised before ridge or the lasso because:

- **A.** the software requires it
- **B.** the penalty compares coefficient sizes, which depend on each variable's units
- **C.** it makes the coefficients significant
- **D.** it removes outliers

A37. *(S5 · medium)* An analyst standardises the predictors on **all** the rows, then runs 5-fold cross-validation. The cross-validated error is:

- **A.** exactly right
- **B.** too pessimistic
- **C.** too optimistic: information from the held-out folds leaked into training
- **D.** unaffected, because standardising changes nothing

A38. *(S5 · medium)* In K-fold cross-validation, $K = 5$ or 10 is the usual choice because:

- **A.** it balances using most of the data in each fit against the cost and instability of a very large K
- **B.** a larger K is forbidden
- **C.** K must divide the sample size
- **D.** a smaller K always gives the lowest error

A39. *(S5 · medium)* After the lasso selects six variables, the analyst refits OLS on them and reports the p-values. These p-values are:

- **A.** valid, because OLS is unbiased
- **B.** too large
- **C.** identical to the lasso's
- **D.** not valid: the same data chose the variables and then tested them

A40. *(S5 · difficult)* With 200 candidate predictors and 60 observations, the lasso can keep at most:

- **A.** 200 variables
- **B.** 60 variables
- **C.** 6 variables
- **D.** 1 variable

Part B — Short answers

Two or three sentences each. Be precise, not lengthy.

Session 2 · Describing a variable

B1. *(S2 · easy)* Define the mean and the median. Which is robust to an extreme value, and why?

B2. *(S2 · easy)* Define the interquartile range. Why is it preferred to the standard deviation for a skewed variable?

B3. *(S2 · easy)* What does a positive skewness tell you about the shape of a distribution, and about the order of its mean and median?

B4. *(S2 · medium)* A variable is symmetric. Can you apply the 68–95–99.7 rule to it? Explain.

Session 3 · Can the regression be trusted?

B5. *(S3 · easy)* What does ordinary least squares minimise? Why squares, rather than the absolute distances?

B6. *(S3 · medium)* Interpret a slope of 0.31 in a regression of GDP growth (in %) on tertiary education (in % of the population) with a control for investment. Write the full sentence.

B7. *(S3 · difficult)* What does "holding the other regressors fixed" mean, according to the Frisch–Waugh–Lovell result?

B8. *(S3 · easy)* Define R^2. Give two things a high R^2 does **not** tell you.

B9. *(S3 · medium)* Name three things you look for in a plot of residuals against fitted values, and what each would tell you about the model.

B10. *(S3 · medium)* AIC and BIC disagree. Which one prefers the smaller model, and why? On what condition may two models be compared at all?

Session 4 · Categorical outcomes

B11. *(S4 · easy)* Give two reasons why a linear regression is a poor model for a 0/1 outcome.

B12. *(S4 · easy)* Define the **odds** of an event. Convert a probability of 0.75 into odds, and say it in words.

B13. *(S4 · medium)* A logit coefficient is 0.26, and $e^{0.26} \approx 1.30$. What is the coefficient measured in, what is 1.30, and what is the **wrong** way to read it?

B14. *(S4 · medium)* Why does a logistic regression have no single "effect on the probability"?

B15. *(S4 · medium)* Distinguish a **Wald** test from a **likelihood-ratio** test. When must you use the second?

B16. *(S4 · difficult)* In an ordinal logit, what are the cutpoints, and what is the proportional-odds assumption? How would you check it?

B17. *(S4 · difficult)* Distinguish **discrimination** from **calibration**. Why does a high AUC not make the predicted probabilities usable?

Session 5 · Regularisation

B18. *(S5 · medium)* Why is stepwise selection unstable, and why is "the data selected these variables" not a finding?

B19. *(S5 · medium)* What does the lasso do that ridge does not, and why? State two of its limitations.

B20. *(S5 · difficult)* Why must the standardisation of the predictors happen **inside** each cross-validation fold?

Session 6 · Curvature

B21. *(S6 · medium)* In $\hat{y} = b_0 + b_1 x + b_2 x^2$, why must you not read b_1 on its own? What does a negative b_2 mean for the effect of x?

Part C — Reading output

Every item contains at least one error, or one tempting wrong reading. Say what it is, why it is wrong, and what you would write instead — in words.

C1. *(S2 · medium)* "Household income in our sample has a mean of €58,000 and a standard deviation of €41,000. By the 68–95–99.7 rule, 95% of households earn between −€24,000 and €140,000."

C2. *(S3 · easy)* A regression of exports on distance reports a coefficient of −0.82 with $p = 0.31$. The analyst writes: *"Distance has no effect on exports."*

C3. *(S3 · medium)* Model A (all 1,200 rows) has AIC 3,410; model B adds a variable missing for 300 country-years and, on the remaining 900 rows, has AIC 2,780. *"Model B is much better."*

C4. *(S4 · medium) "The odds ratio for collateral is 1.40, so borrowers with collateral are 40% more likely to default."*

C5. *(S4 · medium)* A fraud model is reported with **accuracy 0.99**. Fraud occurs in 1% of transactions. The threshold is 0.5, and a missed fraud costs fifty times a false alarm.

C6. *(S5 · difficult)*

```
X = StandardScaler().fit_transform(X_all)          # all rows
lasso = LassoCV(cv=5).fit(X, y)
kept = X_all.columns[lasso.coef_ != 0]
print(smf.ols("y ~ " + " + ".join(kept), data=df).fit().summary())   # p-values reported
```

C7. *(S3 · medium)* A regression of house prices on floor area has $R^2 = 0.91$. The residual plot shows a clear U-shape. The analyst writes: *"With an R^2 of 0.91, the straight line is clearly the right model."*

C8. *(S4 · difficult)* A multinomial logit of programme choice (baseline: *academic*) gives a negative coefficient on the writing score in the *general* equation. A report concludes: *"Better writers are always less likely to choose the general programme."* Yet the predicted probability of *general* rises from 0.19 to 0.23 as the writing score goes from 35 to 45, then falls.

Part D — Interpretation and judgement

A well-argued paragraph. A well-argued refusal earns full marks. Structure: what the number is, what it does not license and why, what evidence would change your answer, what to do meanwhile.

D1. *(S3 · medium)* Across 85 countries, a regression finds that countries with one more year of average schooling export, on average, 6% more high-technology goods ($p < 0.01$, $R^2 = 0.47$). A trade minister wants to announce that extending compulsory schooling by one year "will raise high-technology exports by 6%". Write your reply.

D2. *(S4 · difficult)* A bank's logit finds that applicants from one region have an odds ratio of default of 2.1, significant at 1%. A manager proposes refusing all loans from that region. Write your reply.

D3. *(S5 · medium)* A lasso keeps 6 of 120 candidate indicators to predict export success. A consultant writes: *"The other 114 have no effect on exports."* Write your reply.

D4. *(S4 · difficult)* A vendor sells a credit-scoring model with an AUC of 0.95 and proposes to use its predicted probabilities directly to set each borrower's interest rate. The model has never been checked against observed default rates. Write your reply.

D5. *(S2 · easy)* A report states: *"The average firm in our sample has 250 employees, so the typical firm is medium-sized."* The median is 18 and the skewness is 6.4. What would you correct, and what would you report instead?

Answers: `PRACTICE-ANSWERS.md` · *the exam itself:* `EXAM.md` · *the summary:* `MIDTERM-SUMMARY.pdf`